Learning Objective 3.7: I can apply amplitude tapering (uniform, cosine, Chebyshev, Taylor) to control sidelobe level and predict the pattern trade-off.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a taper off the sliders.** You find the eight Element Gains sliders on the PHASER set to

$$30,\ 60,\ 90,\ 100,\ 100,\ 90,\ 60,\ 30 \text{ percent.}$$

The array is the standard 8-element row, broadside, with the HB100 at boresight. A uniform-taper sweep has already been frozen as the reference trace.

(a) Compute the amplitude sum $\sum a_n$ and the peak drop this taper will show on the sweep plot relative to the frozen uniform trace.

$\sum a_n$ = [ ]

peak drop = [ ]

(b) Compute the taper efficiency η_t for this distribution, as a ratio and in dB.

η_t = [ ]

η_t = [ ]

(c) Your two answers differ by 2.5 dB. State which one the Rectangular plot shows you, and which one belongs in a link budget as the antenna's loss.

(d) The uniform 8-element array has a broadside directivity of about 8.9 dB. What is the directivity of the tapered array?

D = [ ]

2. **Predicting a sweep.** A lab group presses the **Blackman** preset, which sets the eight elements to

$$6,\ 27,\ 66,\ 100,\ 100,\ 66,\ 27,\ 6 \text{ percent},$$

and prepares to sweep. Their frozen uniform reference measured a 13.1° half-power beamwidth with its first sidelobe at -13 dBc, and the sweep's noise floor sits 23 dB below the uniform-taper peak.

(a) Predict the peak drop the Blackman trace will show relative to the frozen uniform trace.

peak drop = ____

(b) The measured Blackman beamwidth is 23.1°. By what factor has the beam broadened, and what physical change in the array produced it?

broadening = ____

(c) Theory puts Blackman's first sidelobe roughly 33 dB below its own peak. What will the group see at that angle on the sweep, and what should they write in the sidelobe column of their table?

(d) A group member argues that Blackman is the better taper because its sidelobes are lower than Hann's. Using only what this measurement can show, evaluate that claim.

3. **A measurement that disagrees.** A student applies the **Hann** preset ($12,\ 43,\ 77,\ 100,\ 100,\ 77,\ 43,\ 12$ percent), predicts a -4.7 dB peak drop from the gain list, and measures -6.8 dB against the frozen uniform trace. The beam is also wider than the 19.5° the expectation table predicts.

   (a) Run the peak-drop formula backwards to find the amplitude sum the array achieved.

   $\sum a_n =$ ☐

   (b) Identify the fault. Which element is misbehaving, and what amplitude is it contributing?

   (c) Compute the taper efficiency of the array as it stands, with that element at zero.

   $\eta_t =$ ☐

   $\eta_t =$ ☐

   (d) Name one feature of the trace itself, other than the peak level, that confirms your diagnosis.

4. **Designing to a specification.** You need a taper that keeps the half-power beamwidth no wider than 17° while pushing the first sidelobe below -20 dBc. With Enforce Symmetric Taper on, you set a linear ramp from 40% at the ends to 91% at the center:

$$40,\ 57,\ 74,\ 91,\ 91,\ 74,\ 57,\ 40 \text{ percent.}$$

The measured sweep returns a 15.5° beamwidth with nothing visible above the noise floor outside the main lobe.

(a) Compute the peak drop and the taper efficiency of this design.

peak drop =

η_t =

(b) The measurement shows nothing above the noise floor outside the main lobe. Explain what you can and cannot conclude about whether the -20 dBc part of the specification is met.

(c) Compare this design against the Hann preset (-4.7 dB peak drop, -1.2 dB taper efficiency, 19.5° beamwidth). State which you would fly on a radar that must resolve two targets 18° apart, and why.

beamwidth margin =

(d) Enforce Symmetric Taper was on for this design. State what would change in the measured pattern if you set the same eight values with the switch off and mistyped one of them.

Documentation: _____